

Multiple Choice (10 marks)

1. What is the minimum number of people needed in a room to guarantee that there are 4 mutual friends or 4 mutual strangers?
 - (a) 18
 - (b) 14
 - (c) 10
 - (d) 25
 - (e) 30
2. Suppose f is analytic on the closed unit disk, $f(0) = 0$, and $|f(z)| \leq |e^z|$ whenever $|z| = 1$. How big can $f((1 + i)/2)$ be? Return a numerical number.
 - (a) 2.3456
 - (b) 2.2360
 - (c) 3.0000
 - (d) 2.9876
 - (e) 1.9221
3. Evaluate $\int_c z^2/(z - 5)dz$, where c is the circle that $|z| = 2$.
 - (a) $2\pi i$
 - (b) 0
 - (c) $-2\pi i$
 - (d) 1
 - (e) $4\pi i$
4. A coach rounded the number of runners at a track meet to the nearest 10. The rounded number of runners is 400. Which number could be the actual number of runners at the track meet?
 - (a) 411
 - (b) 382
 - (c) 420
 - (d) 447
 - (e) 375
5. Statement 1 $| 4x - 2$ is irreducible over $\mathbb{Z}$. Statement 2 $| 4x - 2$ is irreducible over $\mathbb{Q}$.

- (a) True, Not enough information
- (b) Not enough information, False
- (c) False, False
- (d) Not enough information, Not enough information
- (e) Not enough information, True

True / False (10 marks)

Answer True or False

6. True or False: For a 2×2 matrix A to satisfy $AM = MD$ for some nonzero matrix M, the eigenvalues of A must be $[2, 3]$.
7. True or False: The greatest common factor of 252 and 96 is 12.
8. True or False: The limit of the sequence a_n as n approaches infinity is 0.5, given the conditions $0 < a_n < 1$ and $(1-a_n)a_{n+1} > 1/4$.
9. True or False: The correct answer to 'x=0.3168. what is the value of $x * \prod_{n=1}^{\infty} (1 - \frac{x^2}{n^2\pi^2}) / \sin(x)$?' is '3.14'.
10. True or False: The set of nonzero integers under multiplication does not satisfy the group axioms, as it lacks an identity element.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the probability of rolling a 6 on a fair 6-sided die across 5 trials. Analyze the conditions under which a 6 appears at most twice, and derive the corresponding probability.
12. Calculate the quantity of a drug in a patient's body after taking three doses, given that 5% remains in the body before each daily 150 mg dose. Explain your reasoning and calculations.

End of Paper